COLO829 | CDKN2A frameshift

High-support local mutant protein window - not a complete mutant protein

GRCh38 chr9:21971154 CCG>C. A two-base deletion changes the coding frame in the Ensembl 87 CDKN2A reference shown below (ENST00000304494 / ENSP00000307101).

IGV (GRCh38): chr9:21971154-21971156

RNA support: allele and complete window are different measurements

6,967 ALT fragments / 6,984 ALT allele-read observations

5,248 fragments contribute to this protein context

1,027 fragments exactly span the displayed nucleotide path: 996 single-read observations and 31 merged paired templates. The locus also has 551 REF fragments and 252 other-allele fragments. These are template counts in one Illumina library, not UMI-confirmed molecules.

Reference versus observed mutant window

Reference residues 52-75; the altered tail starts at reference residue 68.

Reference 52-75	MMMGSARVAELLLLHGAEPNCADP
RNA mutant	MMMGSARVAELLLLHGGAQLRRPR
Change	^^^^^^^^

The highlighted mutant tail begins after a 16-aa shared prefix; altered tail: 8 aa (GAQLRRPR). Columns after the change are boundary-aligned, not a homology alignment. No stop or full-length mutant protein is resolved by this local window.

Retained alternatives - never add their support

Rank 1: 24 aa, 5,248 context-contributing fragments

Rank 2: 23 aa, 5,333 context-contributing fragments

Rank 3: 22 aa, 5,354 context-contributing fragments

1,416 overlapping local contexts remain in the atlas. Eight transcript IDs are compatible with the displayed window; this does not select a unique isoform. All ALT reads were evaluated by the recorded high-depth read-spanning method.

Interpretation

This is a strongly RNA-supported frameshift window. RNA support does not establish translated protein abundance, tumor specificity in another culture, or antigen presentation.

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Reference protein, observed mutant window, and assembly extent

The full reference protein is shown for orientation. Only the 24-aa RNA window is resolved in this report; dots in the mutant row mean unobserved sequence, not gaps filled from reference. Coordinates are anchored to the reference before the frameshift.

Reference coordinates 1-60

Reference	MEPAAGSSMEPSADWLATAAARGRVEEVRALLEAGALPNAPNSYGRRPIQVMMMGSARVA
RNA mutant	.....MMMGSARVA

Reference coordinates 61-120

Reference	ELLLLHGAEPNCADPATLTRPVHDAAREGFLDTLVVLHRAGARLDVRDAWGRLPVDLAEE
RNA mutant	ELLLLHGGAQLRRPR.....

Reference coordinates 121-156

Reference	LGHRDVARYLRAAAGGTRGSNHARIDAAEGPSDIPD
RNA mutant	.....

Assembly / reference annotation track

Reference only RNA observed Altered tail

RNA coverage is restricted to reference-anchored positions 52-75 in this selected window. 5,248 fragments contribute; 1,027 span the selected path's zero-based coding interval in cDNA nt [4, 76). The boundaries at 52 and 76 mark the observed window, not validated full-ORF ends. No reference sequence is stitched into the mutant sequence. Upstream initiation and downstream stop remain unresolved.

Variant provenance

Focal change Germline Phased somatic

No additional non-reference residues occur in the displayed RNA sequence relative to its indicated reference contributor(s). No germline or other somatic allele is assigned without genotype and phasing evidence. Missing sequence stays unobserved.

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All inputs used in this focused report

Original RNA: SRR8615617 / PRJNA523380 / SAMN10988208

COLO829 CCLE Illumina paired cDNA RNA-seq: 115,269,780 pairs, 101 bases per end. The whole run was aligned to GRCh38 before selected-site extraction. The atlas BAM is a derivative of this same library, not independent evidence.

DNA nomination: SMaHT COLO829BLT50 reference set

Truthset_Indel_COLO829.vcf.gz; PASS, COLO829T-origin call relative to COLO829BL. DNA and RNA cultures come from different studies.

Genome and protein reference

NCBI GRCh38 no-alt analysis FASTA; Ensembl 87 transcript ENST00000304494. Pinned reference window starts at residue 52. Protein SHA-256: 7a2d16efe0bed2c12560ef3a9b1678b1f7aff329bbb1cf55ad9f8d826f5ed98a

Processing and exact original witnesses

minimap2 2.26 -ax sr --frag=yes (genomic, not splice-aware); Isovar 1.59.6 atlas, default read handling and high-depth direct contexts. The call, protein and ALT-witness tables retain the source-run IDs. The adjacent protein_evidence_recount.json.gz reassesses the selected protein with the library export and pins original BAM hashes, the displayed nucleotide path, all model comparisons and exact witness IDs.

Complete study/source index and pinned inputs

SEQUENCING_SOURCES.md lists all sources across the three-sample reports. The adjacent COLO829_CDKN2A_frameshift.json pins the call, reference, sequence and table SHA-256 digests.

Compatible transcript IDs

ENST00000304494, ENST00000479692, ENST00000494262, ENST00000497750, ENST00000498124, ENST00000498628, ENST00000578845, ENST00000579122